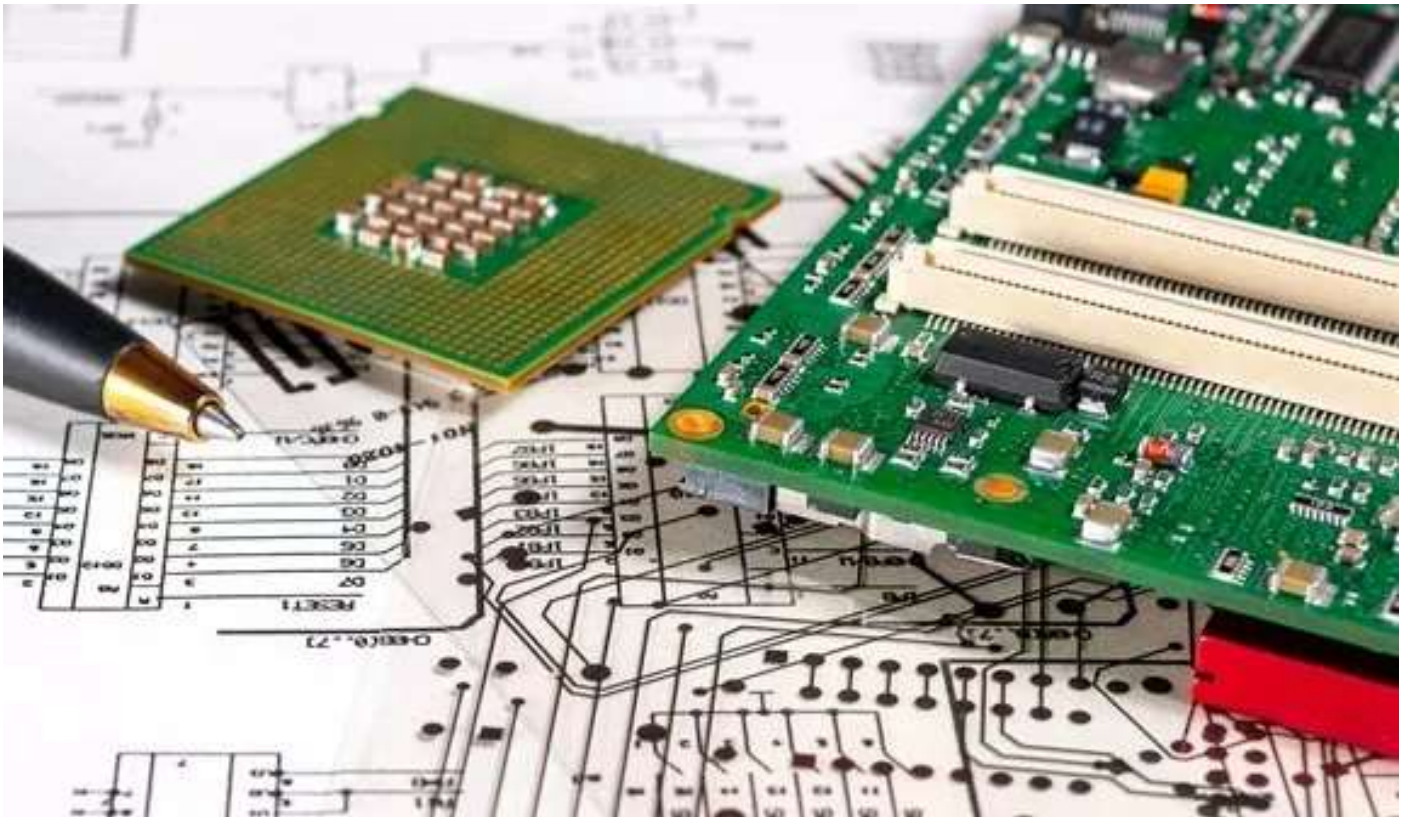




The realization of the different parts can be done independently.

I- Design a system allowing the autonomous navigation of the robot in the picking path.

- 1- Develop a SLAM mapping algorithm under ROS to construct a 10m x 5m simulated agricultural scene map.
- 2- Combined with point 3, the “zero spin radius” characteristic of the Mecanum wheel (spin radius ≤ 0 mm).
- 3- Design an A* path planning algorithm suitable for the crop row spacing.
- 4- Realize the robot’s autonomous cruise along the planting row.
- 5- Ultrasonic obstacle avoidance
- 6- Precision of docking at picking points.
- 7- Adapt the navigation to the narrow working space of agricultural scenarios such as greenhouses.

II. Development of Crop Recognition and Maturity Judgment Module

- 1- Collect data on simulated crops commonly used in competitions (such as strawberries and tomatoes).
- 2- Construct a crop image dataset with "unripe-ripe-overripe" labels.
- 3- Use YOLOv5s/MobileNet lightweight deep learning models to develop crop recognition and maturity classification algorithms.
- 4- Improve agricultural scene adaptability through transfer learning (pre-trained based on COCO dataset), output 2D coordinates of RGB images and 3D depth positions of mature crops, ensuring recognition accuracy $\geq 92\%$ and single-frame processing time $\leq 0.3s$.
- 5- Combined with the 60° field of view of the camera module.

III. Precision Picking Control of Robotic Arm and Adaptive Gripper

1. Complete the hand-eye calibration of the 3-4 DOF robot arm based on ROS hand-eye calibration tools.
2. Establish a coordinate mapping model of "depth camera-robotic arm base coordinate-gripper end".
3. Convert the 3D position of crops into robotic arm motion commands.
4. Develop picking trajectory planning logic through Python programming interface.
5. Control the robotic arm to act according to the process of "approaching crops → adaptive gripper opening → precise clamping (automatic force adjustment according to crop diameter) → lifting off the plant → transferring to the storage box".
6. Optimize the gripper control strategy: adjust the clamping force through pressure sensor feedback for different crops such as strawberries (diameter 2-4cm) and tomatoes (4- 6cm) to avoid crop damage or falling.

IV. System Integration and Performance Optimization

- 1- Integrate the voice control function (based on TTL serial voice module, 5V power supply), support voice commands such as "start picking", "pause picking", and "switch crop type" (Response time $\leq 1s$) to adapt to the convenient operation needs of agricultural scenarios.
- 2- Increase the picking volume per unit time to ≥ 15 pieces/minute.
- 3- Build a simulated agricultural scene test platform
- 4- Monitor the stability of the robotic arm motor PWM signal (ripple $\leq 100mV$), and system power consumption (full-load cruise $\leq 15W$) with oscilloscopes and multimeters.
- 5- Optimize the lithium battery power supply strategy (working time $\geq 4h$)
- 6- Test the system stability at 0~40°C and 5%-80% humidity to ensure trouble-free operation $\geq 8h$.

V- design of the assembly of the car and the robot arm

- 1- Learn the basics of modeling with Onescape.
- 2- Learn how to import objects and manipulate them on Oneshape, to integrate the car and the robot arm on Oneshape for the complete design.
- 3- Have the resulting structure of the design printed.
- 4- Assemble the different parts (the car and the robot arm).
- 5- Final Test.